CS5491 AI ・Module E: Bayesian Networks Inference & Sampling
Week12 Tutorial 优先：BN 表示・D-separation ・Exact Inference ・Variable Elimination ・Sampling

0. 题型识别：看到关键词就套模板

Question keywords	考场动作 / What to do
draw BN / causes	画 DAG：root prior；child 连 direct causes
joint distribution	$\prod_i P(X_i \mid Pa_i)$，一个节点一个因子
number of parameters	Bernoulli node with m parents needs 2^m params
independent / d-separated	列 paths，按 chain/fork/collider 判断 blocked
$P(Q \mid e)$ exact	hidden 求和，再 normalize
factor / join / eliminate	join= 乘；eliminate=$\sum_X$ 消变量
elimination order	比最大中间 factor 变量数 / entries
given samples	prior/rejection: 普通 count
weighted samples	likelihood weighting: weighted count
resample one variable	Gibbs: Markov blanket score 再 normalize

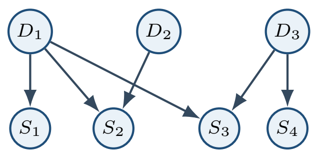
1. BN 基础：Graph + CPTs

Bayesian Network = directed acyclic graph (DAG) + local conditional probability tables (CPTs).
A BN represents a joint distribution as a product of local conditional distributions.

$$P(x_1,\dots,x_n)=\prod_i P(x_i \mid Parents(X_i))$$

完整 assignment 概率：把每个节点对应的 CPT entry 乘起来。Root 用 prior，有父节点用 conditional。

Tutorial Network Basics：医生诊断



$$\begin{aligned}&P(D_1,D_2,D_3,S_1,S_2,S_3,S_4)\\&=P(D_1)P(D_2)P(D_3)P(S_1\mid D_1)P(S_2\mid D_1,D_2)\\&\ P(S_3\mid D_1,D_3)P(S_4\mid D_3)\end{aligned}$$

参数数：Bernoulli 变量，每个 parent assignment 只需 1 个自由参数；m 个 binary parents $\Rightarrow 2^m$ params。一般公式：若 X 有 r 个值，parents assignments 数为 M，参数数 $=(r-1)M$。上图：$1+1+1+2+4+4+2=15$。无条件独立、全连接：$1+2+4+8+16+32+64=127=2^7-1$。

2. D-Separation：Tutorial T/F 题模板

结构	不观察中间点	观察中间点
Chain $X\to Y\to Z$	active	blocked
Fork $X\leftarrow Y\to Z$	active	blocked
Collider $X\to Y\leftarrow Z$	blocked	active if Y or descendant observed

口诀：链/叉，看见中间点就堵住；撞，平时堵住，看见撞点或后代就打开。
List every path. If all paths are blocked, variables are conditionally independent.

Tutorial 医生图：观察 S_4 学到什么？

无其他 evidence：S_4 只直接连 D_3，关于 D_1,D_2 的路径被 S_3/S_2 collider 挡住，所以只学到 D_3。
若已知 $S_3{=}T$: $D_1\to S_3\leftarrow D_3\to S_4$ 的 collider 被打开，观察 S_4 也会给 D_1 信息（explaining away）；仍不给 D_2 信息。

3. Week12 D-Separation 答案速查

Question	Ans	Why / 可写理由
Party ⊥ Success ∣ HW?	False	仍有未堵路径，如 Party–HW–Smart–Project–Success；HW 被观察也会打开 collider。
Creative ⊥ Happy ∣ Mac?	False	Creative–Project–Success–Happy 仍 active。
Party ⊥ Smart ∣ Success?	False	Party $\to$ HW $\leftarrow$ Smart 中 HW 是 collider；observed Success 是 HW 的 descendant，所以打开该路径。
Party ⊥ Creative ∣ Happy?	False	Happy 是 collider，被观察后打开 Party–Happy–Mac–Creative 等路径。
Party ⊥ Creative ∣ Success, Project, Smart?	True	到 Creative 的主要路径被 observed non-collider Smart/Project 或未打开 collider 堵住。

写 T/F 不要只写答案：至少写一条 active path 证明 False；或写 all paths blocked 证明 True。

4. Exact Inference：条件概率题

Query Q：竖线左边要求的变量；**Evidence** e：竖线右边已知变量；**Hidden** H：其余变量。

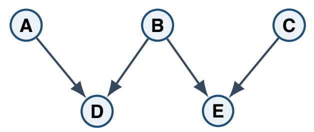
$$P(Q\mid e)=\alpha P(Q,e)=normalize\left(\sum_H\prod_i P(x_i\mid Pa_i)\right)$$

Enumeration 步骤:

1. 写 joint factorization。
2. 固定 evidence。
3. 对 hidden variables 求和。
4. 得到 query 每个取值的 unnormalized score。
5. Normalize：每个 score 除以总和。

Select entries consistent with evidence, sum out hidden variables, then normalize over the query variable.

Tutorial Inference BN



$$P(A,B,C,D,E)=P(A)P(B)P(C)P(D\mid A,B)P(E\mid B,C)$$

A	$P(A=T)$	$\mid$ B	$P(B=T)$	C, $P(C=T)$
	0.2	$\mid$	0.5	0.8

A	B	$P(D=T)$
F	F	.9
F	T	.6
T	F	.5
T	T	.1

B	C	$P(E=T)$
F	F	.2
F	T	.4
T	F	.8
T	T	.3

Q1: $P(A=F,B=F,C=F,D=F,E=F)$

$$0.8\cdot0.5\cdot0.2\cdot(1-0.9)\cdot(1-0.2)=0.0064$$

模板：full assignment 不 normalize，直接按 joint 乘。

Q2: $P(A=F\mid B=T,C=T,D=T,E=T)$

$$s_F=0.8\cdot0.5\cdot0.8\cdot0.6\cdot0.3=0.0576$$

$$s_T=0.2\cdot0.5\cdot0.8\cdot0.1\cdot0.3=0.0024$$

$$P(A=F\mid B,C,D,E=T)=\frac{s_F}{s_F+s_T}=0.96$$

口诀：只问 A，就算 $A=F$ 和 $A=T$ 两个 score，再 normalize。

5. Factor / Join / Eliminate

Factor = probability table over some variables. Evidence 被固定后，该变量从 factor 的维度里消失。

- **Join factors**: compatible entries pointwise multiply; variables union.
  例：$P(R)$ join $P(T\mid R)$ gives $f(R,T)=P(R)P(T\mid R)$。
- **Eliminate / sum out**: 对变量求和，factor 变小。
  若 $f(X,Y)$，消 X 得 $g(Y)=\sum_X f(X,Y)$。

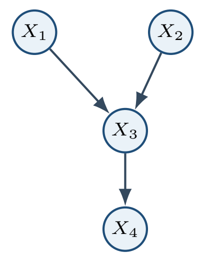
6. Variable Elimination 通用算法

目的：精确计算 $P(Q\mid e)$ 或 $P(Q)$，但不要先构造完整 joint。
VE interleaves factor joining and marginalization to avoid building the full joint table.

1. Start with initial factors = all local CPTs.
2. Instantiate evidence: fix evidence values in factors.
3. For each hidden variable H in chosen order: join all factors mentioning H; eliminate H by summing it out; put new factor back.
4. Join remaining factors.
5. Normalize if computing a conditional/posterior $P(Q\mid e)$; for pure marginal $P(Q)$ it is often already normalized.

Order matters：中间 factor 变量越多，表越大。Bernoulli/k-valued 下，m 个变量的 factor 大小约 k^m。

Tutorial VE 网络与目标



$$P(x_1,x_2,x_3,x_4)=P(x_1)P(x_2)P(x_3\mid x_1,x_2)P(x_4\mid x_3)$$

$$P(x_4)=\sum_{x_1}\sum_{x_2}\sum_{x_3}P(x_1)P(x_2)P(x_3\mid x_1,x_2)P(x_4\mid x_3)$$

Good order: eliminate X_1 first

$$g_1(x_3,x_2)=\sum_{x_1}P(x_1)P(x_3\mid x_1,x_2)$$

$$g_2(x_3)=\sum_{x_2}P(x_2)g_1(x_3,x_2),\quad P(x_4)=\sum_{x_3}P(x_4\mid x_3)g_2(x_3)$$

Complexity: largest intermediate $g_1(x_3,x_2)$ has k^2 entries; total $O(k^3)$.

Bad order: eliminate X_3 first

$$g(x_1,x_2,x_4)=\sum_{x_3}P(x_3\mid x_1,x_2)P(x_4\mid x_3)$$

Largest intermediate has k^3 entries; total $O(k^4)$。所以 X_1 first preferable。

7. 英文短答直接抄

BN: A Bayesian network is a DAG with one CPT per node. It factorizes the joint distribution as $\prod_i P(X_i\mid Parents(X_i))$.
Enumeration: It computes $P(Q\mid e)$ by selecting assignments consistent with evidence, summing out hidden variables, and normalizing over Q.
VE: Variable elimination joins factors mentioning each hidden variable and sums that variable out. For a posterior query it normalizes at the end. The elimination order affects intermediate factor size.

Module E Back: Tutorial VE 数值题 + Sampling 套题模板

Prior ・Rejection ・Likelihood Weighting ・Gibbs ・Markov Blanket ・考场检查

8. Tutorial VE 数值题：Eliminate X_1 First

x_1	$P(x_1)$	x_2	$P(x_2)$
0	.3	0	.6
1	.7	1	.4

x_3	x_1	x_2	$P(x_3 \mid x_1, x_2)$
0	0	0	.5
0	0	1	.2
0	1	0	.9
0	1	1	.5
1	0	0	.5
1	0	1	.8
1	1	0	.1
1	1	1	.5

x_4	x_3	$P(x_4 \mid x_3)$
0	0	.7
0	1	.1
1	0	.3
1	1	.9

Step 1: compute $g_1(x_3, x_2)$

$$g_1(x_3, x_2) = \sum_{x_1} P(x_1)P(x_3 \mid x_1, x_2)$$

x_3	x_2	$g_1(x_3, x_2)$
0	0	$.3(.5) + .7(.9) = .78$
1	0	$.3(.5) + .7(.1) = .22$
0	1	$.3(.2) + .7(.5) = .41$
1	1	$.3(.8) + .7(.5) = .59$

Step 2: compute $g_2(x_3)$

$$g_2(x_3) = \sum_{x_2} P(x_2)g_1(x_3, x_2)$$

$$g_2(0) = .6(.78) + .4(.41) = .632, \qquad g_2(1) = .6(.22) + .4(.59) = .368$$

Step 3: compute $P(x_4)$

$$P(x_4) = \sum_{x_3} P(x_4 \mid x_3)g_2(x_3)$$

$$P(x_4 = 0) = .7(.632) + .1(.368) = .4792$$

$$P(x_4 = 1) = .3(.632) + .9(.368) = .5208$$

9. Tutorial VE：Eliminate X_3 First

$$g(x_1, x_2, x_4) = \sum_{x_3} P(x_3 \mid x_1, x_2)P(x_4 \mid x_3)$$

x_1	x_2	x_4	$g(x_1, x_2, x_4)$
0	0	0	$.5(.7) + .5(.1) = .40$
0	0	1	$.5(.3) + .5(.9) = .60$
0	1	0	$.2(.7) + .8(.1) = .22$
0	1	1	$.2(.3) + .8(.9) = .78$
1	0	0	$.9(.7) + .1(.1) = .64$
1	0	1	$.9(.3) + .1(.9) = .36$
1	1	0	$.5(.7) + .5(.1) = .40$
1	1	1	$.5(.3) + .5(.9) = .60$

后续继续消 X_2, X_1 会得到同样 final $P(x_4)$，但中间 factor 更大。

Sum-product calculations

X_1 **first**: $p(x_3 \mid x_2)$ has 4 calculations; $p(x_3)$ has 2; $p(x_4)$ has 2. Total $4 + 2 + 2 = 8$.
X_3 **first**: $p(x_4 \mid x_1, x_2)$ has 8; $p(x_4 \mid x_1)$ has 4; $p(x_4)$ has 2. Total $8 + 4 + 2 = 14$.
X_1 **first is preferable** because it creates smaller intermediate factors.

10. Sampling 大局观

为什么 sampling? Exact inference may be expensive because intermediate factors can be huge. Sampling approximates probabilities using samples.
Sampling estimates probabilities by sample frequencies or weighted frequencies. More samples generally give a better approximation.

Method	Query	核心动作	Count
Prior	$P(Q)$	topological sample all	frequency
Rejection	$P(Q \mid e)$	prior samples, reject non-e	accepted frequency
Likelihood W.	$P(Q \mid e)$	fix evidence, weight samples	weighted frequency
Gibbs	$P(Q \mid e)$	fix evidence, resample variables	state frequency

11. Prior Sampling：无 Evidence

用在： $P(Q)$ 或估计 prior/marginal。
Algorithm:

1. Order variables topologically: parents before children.
2. Sample each X_i from $P(X_i \mid Parents(X_i))$.
3. Repeat N times.
4. Estimate by counting frequencies.

$$S_{PS}(x_1, \ldots, x_n) = \prod_i P(x_i \mid Pa_i) = P(x_1, \ldots, x_n)$$

Count 模板： 5 个 samples 中 $+w$ 有 4 个，$-w$ 有 1 个，则 $\hat{P}(W) = \langle +w : .8, -w : .2 \rangle$。

12. Rejection Sampling：先采样再扔

用在： $P(Q \mid e)$，给 ordinary samples。

1. Generate complete prior samples.
2. Reject samples inconsistent with evidence e.
3. Count Q values among accepted samples.
4. Normalize.

$$\hat{P}(Q = q \mid e) = \frac{\#(Q = q \text{ and } e)}{\#(e)}$$

例： 求 $P(R = T \mid W = T)$。保留 $W = T$ 的 samples；其中数 $R = T$ 的比例。若保留 1,2,4 且其中 1,4 为 $R = T$，答案 2/3。
缺点： evidence rare 时大多数样本被 rejected，效率低。
Inefficient when evidence has low probability because most generated samples are discarded.

13. Likelihood Weighting：固定 Evidence，加权

核心区别： Evidence variables are fixed, not sampled. No sample is rejected.

Algorithm for $P(Q \mid e)$
Initialize $w = 1$. In topological order: if X_i is evidence, fix $X_i = e_i$ and set $w \leftarrow w \cdot P(e_i \mid Parents(E_i))$; otherwise sample X_i from its CPT. Add weight w to the count of the sampled query value. Repeat and normalize.

$$S_{WS}(z, e) = \prod_i P(z_i \mid Pa_i), \qquad w(z, e) = \prod_i P(e_i \mid Pa_i)$$

$$S_{WS}(z, e)w(z, e) = P(z, e)$$

Weighted count:

sample	Q	weight
1	T	.9
2	F	.2
3	T	.8
4	F	.1

$$\hat{P}(Q = T \mid e) = \frac{.9 + .8}{.9 + .8 + .2 + .1} = 0.85$$

缺点： downstream evidence 不会直接改变 upstream sampling，只靠 weight 修正。
Upstream variables are not sampled conditioned on downstream evidence.

14. Gibbs Sampling：固定 Evidence，反复重采样

用在： $P(Q \mid e)$，希望 evidence 影响上下游。

1. Fix evidence variables.
2. Randomly initialize all non-evidence variables.
3. Repeatedly choose one non-evidence variable X.
4. Resample X from $P(X \mid rest)$.
5. Estimate $P(Q \mid e)$ from later states.

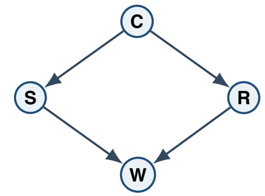
Gibbs sampling is MCMC. In the limit, it converges to the correct posterior distribution.

Markov Blanket 重采样公式

$$P(X \mid rest) \propto P(X \mid Parents(X)) \prod_{Y \in Children(X)} P(Y \mid Parents(Y))$$

只看包含 X 的 CPT： X 自己的 CPT + X 的 children 的 CPT。其他项 normalize 时抵消。

课件例：Sprinkler 网络 resample S



求 $P(S \mid +c, +r, -w)$:

$$score(+s) = P(+s \mid +c)P(-w \mid +s, +r)$$

$$score(-s) = P(-s \mid +c)P(-w \mid -s, +r)$$

$$P(+s \mid +c, +r, -w) = \frac{score(+s)}{score(+s) + score(-s)}$$

15. Rejection vs Likelihood Weighting 高频比较

	Rejection	Likelihood Weighting
Evidence	sampled first	fixed to observed value
Bad sample	discard	keep with weight
Count	accepted frequency	weighted frequency
Worst case	rare evidence	upstream not directly affected

Rejection sampling discards samples inconsistent with evidence, so it is inefficient when evidence is rare. Likelihood weighting fixes evidence variables and weights each sample by the likelihood of the evidence, so every sample contributes.

16. 考场检查清单

1. 有没有 evidence？没有多半 prior/marginal；有就是 $P(Q \mid e)$。
2. 给 CPT 还是 samples？CPT 可 exact/VE；samples 用 count；weighted samples 用 weighted count。
3. 条件概率有没有 normalize？posterior 要；full joint assignment 不要。
4. Hidden variables 是谁？All variables − query − evidence。
5. VE order 问哪个好？比较最大 intermediate factor 的变量数/entries。
6. Bernoulli 参数数：node with m parents needs 2^m free parameters。
7. D-separation：False 写 active path；True 写 all paths blocked。

17. 考场落笔模板：按空填

Exact inference: Query = ________, evidence = ________, hidden = all others. Compute one score for each query value by multiplying CPT entries and summing hidden variables; normalize scores.
VE order: Eliminating _____ creates a factor over ___________, size k^{-}. The smaller largest factor is preferable.
Sampling count: ordinary samples $\Rightarrow$ count rows; evidence query $\Rightarrow$ keep evidence rows first; weighted samples $\Rightarrow$ sum weights, not rows.
Gibbs score: write one unnormalized score per value of resampled variable, using only its CPT and its children's CPTs, then normalize.